

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Civil Examination

May 2011

CL201 SURVEYING

Date: 17.5.2011, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Enlist the methods of plane table surveying. Describe the method of orienting plane table by backsighting. 04
- (b) What is the application of total station in modern surveying? 03
- Q - 2 (a) Explain the process of setting out of small. 04
- (b) What is meant by degree of curve? Derive its relationship with the radius of curve. 04
- (c) Two straight lines intersect at a chainage of 1250.50m, the angle of intersection is 80° . If the degree of the curve is 5° , by using 30m chain, determine the following 06
- a) Radius of curve
 - b) Tangent distance
 - c) Length of curve
 - d) Length of long chord
 - e) Chainage of point of curvature
 - f) Apex distance and the mid-ordinate

OR

- Q - 2 (a) Draw a neat sketch of simple circular curve and show following elements. 04
- a) Long chord
 - b) Point of commencement
 - c) Mid-ordinate
 - d) Rear tangent
- (b) What is the use of slope rail and boning rod in setting out works. 04
- (c) Two straights AB & BC intersect at a chainage of 3450m. The angle of intersection is 130° . It is required to set out 6° simple circular curve to connect the straights. Calculate all the data necessary to set out the curve by method of **offset from chord produced** with an interval of 30m. 06
- Q - 3 (a) Define the following 02
- a) Apex distance
 - b) Fiducial edge of the alidade
 - c) Compound curve
 - d) U- fork
- (b) What are the advantages and disadvantages of plane table survey? 05
- (c) Discuss in details the steps involved in setting out centre line of tunnel 04
- (d) Describe briefly salient features of total station. 03

OR

- Q - 3 (a) Define the following 02
- a) Point of tangency
 - b) Vertical curve
 - c) Trough compass
 - d) Johnson's table
- (b) What are the methods of plane tabling? Explain the method of traversing. 04

P.T.O

OR

- Q – 6 (a)** What is hydrographic survey? What are the main purposes for which hydrographic surveying is carried out? **04**
- (b)** Define the following terms of vernier transit theodolite **05**
- a) upper plate
 - b) transiting or plunging
 - c) face right
 - d) verticle circle
 - e) line of collimation
- (c)** Prepared a Gale's traverse to adjust the closing error of the closed traverse ABCDA for the following data: **05**

Line	Length (M)	Bearing
AB	110.00	110°00'
BC	80.00	170°00'
CD	95.00	250°00'
DA	160.00	350°00'
